

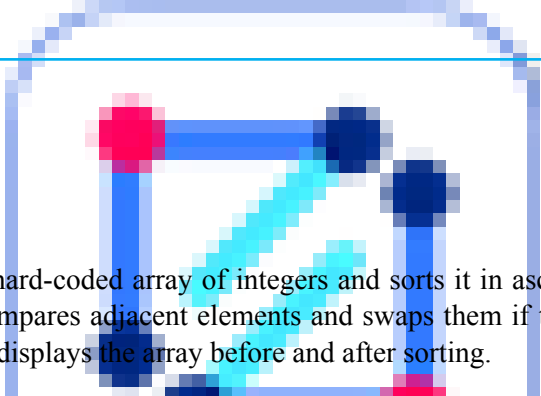


<u>Name</u>	<i><u>Muddassir Ali Siddiqui</u></i>
<u>Instructor</u>	<i><u>Sir Junaid</u></i>
<u>Date</u>	<i><u>22nd Sep 2025</u></i>

1. In-Lab Tasks:

i. Task 1:

I wrote a shell script that stores a hard-coded array of integers and sorts it in ascending order using the Bubble Sort algorithm. The script repeatedly compares adjacent elements and swaps them if they are out of order until the entire array is sorted. The terminal output displays the array before and after sorting.



```
[cc@ncdc-0063 lab27]$ touch bubble_sort.sh
[cc@ncdc-0063 lab27]$ nano bubble_sort.sh
[cc@ncdc-0063 lab27]$ ./devdojo.sh
bubble_sort      devdojo.sh
bubble_sort.sh   Shell Scripting Lab 27.pdf
[cc@ncdc-0063 lab27]$ nano bubble_sort.sh
[cc@ncdc-0063 lab27]$ bash
bubble_sort.sh   Shell Scripting Lab 27.pdf
devdojo.sh
[cc@ncdc-0063 lab27]$ bash bubble_sort.sh
Original array: 64 34 25 12 22 11 90 5 77 30
Array length: 10

Final sorted array: 5 11 12 22 25 30 34 64 77 90

Bubble sort completed successfully!
[cc@ncdc-0063 lab27]$
```

ii. Task 2:

This script prompts the user for a desired password length and then generates a random string containing uppercase and lowercase letters, digits, and special characters. It ensures true randomness and prints the generated password to the terminal.

```
[cc@ncdc-0063 lab27]$ touch rand_pass_gen.sh
[cc@ncdc-0063 lab27]$ nano rand_pass_gen.sh
```

```
[cc@ncdc-0063 lab27]$ nano rand_pass_gen.sh
[cc@ncdc-0063 lab27]$ bash rand_pass_gen.sh
=====
Welcome to Random Password Generator
=====

Enter desired password length (minimum 4, maximum 20): 6

Generating password of length 6...

Generated Password: ]:m_|v
```

Scripting

```
Password Analysis:
Contains lowercase: Yes
Contains uppercase: No
Contains digits: No
Contains special chars: Yes
Strength: Medium

Password generation completed successfully!

[cc@ncdc-0063 lab27]$
```

iii. Task 3:

I developed a script that mimics basic Git functionality with three functions: **init**, **add**, and **status**.

- **init** is run once at the beginning to create a hidden backup folder, setting up the directory for version tracking.
- **add** copies any new or updated files from the working directory into this hidden folder, effectively staging them.
- **status** compares the current working directory with the hidden backup to show which files are new, modified, or deleted, giving a Git-like status report.

```
● @Ali-975 → /workspaces/shell-scripting (main) $ bash git_init_status.sh
Custom Git Simulator

Usage: git_init_status.sh [command]

Available commands: init, status, add, help
Run 'git_init_status.sh help' for more information
● @Ali-975 → /workspaces/shell-scripting (main) $ git_init_status.sh help
bash: git_init_status.sh: command not found
● @Ali-975 → /workspaces/shell-scripting (main) $ bash git_init_status.sh status
Checking repository status...

Error: Not a git repository (or any of the parent directories)
Run 'git_init_status.sh init' to initialize a repository
○ @Ali-975 → /workspaces/shell-scripting (main) $
```

Scripting

```
• @Ali-975 → /workspaces/shell-scripting (main) $ bash git_init_status.sh init
Initializing custom git repository...
Creating initial backup...
Custom git repository initialized successfully in /workspaces/shell-scripting
Backup created in .mygit_backup directory
• @Ali-975 → /workspaces/shell-scripting (main) $ bash git_init_status.sh status
Checking repository status...

On branch master

No changes detected.
Working tree clean
• @Ali-975 → /workspaces/shell-scripting (main) $ bash git_init_status.sh add
Updating backup with current changes...
Changes staged successfully!
• @Ali-975 → /workspaces/shell-scripting (main) $ bash git_init_status.sh status
Checking repository status...

On branch master

No changes detected.
Working tree clean
○ @Ali-975 → /workspaces/shell-scripting (main) $ █
```

```
• @Ali-975 → /workspaces/shell-scripting (main) $ ./task_3.sh status
Checking repository status...

On branch master

Untracked files:
  (use "git add <file>..." to include in what will be committed)

    readme.md
    task_1.sh
    task_2.sh
    task_3.sh
    task_4.sh
    task_5.sh

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git checkout -- <file>..." to discard changes in working directory)

    deleted:    bubble_sort.sh
    deleted:    git_init_status.sh
    deleted:    rand_pwd_gen.sh

Summary: 6 new, 0 modified, 3 deleted
❖ @Ali-975 → /workspaces/shell-scripting (main) $ █
```

Scripting

```
Summary: 0 new, 0 modified, 0 deleted
• @Ali-975 → /workspaces/shell-scripting (main) $ ./task_3.sh add
Updating backup with current changes...
Changes staged successfully!
• @Ali-975 → /workspaces/shell-scripting (main) $ ./task_3.sh status
Checking repository status...

On branch master

No changes detected.
Working tree clean
❖ @Ali-975 → /workspaces/shell-scripting (main) $
```

iv. Task 4:

This script scans a **files/** directory, creates subfolders for **.pdf**, **.c**, and **.h** files, and moves files into their respective subdirectories. After segregation, it compresses each subdirectory into a **.zip** archive and places the archives back into the **files/** folder for organized storage.

```
• @Ali-975 → /workspaces/shell-scripting (main) $ ./task_4.sh
=====
File Type Segregation and Archiving
=====

Creating files directory...
The files directory is empty.
Would you like to create sample files for demonstration? (y/n): y
Creating sample files for demonstration...
Sample files created successfully!

Initial contents of files:
total 24
drwxrwxrwx+ 2 codespace codespace 4096 Sep 23 06:37 .
drwxrwxrwx+ 5 codespace root      4096 Sep 23 06:37 ..
-rw-rw-rw-  1 codespace codespace    0 Sep 23 06:37 data.json
-rw-rw-rw-  1 codespace codespace    0 Sep 23 06:37 document1.pdf
-rw-rw-rw-  1 codespace codespace   94 Sep 23 06:37 hello.c
-rw-rw-rw-  1 codespace codespace   61 Sep 23 06:37 main.h
-rw-rw-rw-  1 codespace codespace    0 Sep 23 06:37 manual.pdf
-rw-rw-rw-  1 codespace codespace    0 Sep 23 06:37 readme.txt
-rw-rw-rw-  1 codespace codespace    0 Sep 23 06:37 report.pdf
-rw-rw-rw-  1 codespace codespace   81 Sep 23 06:37 utils.c
-rw-rw-rw-  1 codespace codespace   69 Sep 23 06:37 utils.h
```

```
Creating subdirectories...
Created directory: files/pdf_files
Created directory: files/c_files
Created directory: files/h_files

File segregation completed!
PDF files moved:
Processing .pdf files...
Moving: document1.pdf -> files/pdf_files/
Moving: manual.pdf -> files/pdf_files/
Moving: report.pdf -> files/pdf_files/
Moved 3 .pdf file(s)
3
C files moved:
Processing .c files...
Moving: hello.c -> files/c_files/
Moving: utils.c -> files/c_files/
Moved 2 .c file(s)
2
Header files moved:
Processing .h files...
Moving: main.h -> files/h_files/
Moving: utils.h -> files/h_files/
Moved 2 .h file(s)
2
```

```
Creating archives...

Creating archive: pdf_files.zip
Archive created successfully: files/pdf_files.zip
Archive contents:
 0  2025-09-23 06:37  pdf_files/
 0  2025-09-23 06:37  pdf_files/manual.pdf
 0  2025-09-23 06:37  pdf_files/report.pdf
 0  2025-09-23 06:37  pdf_files/document1.pdf

Creating archive: c_files.zip
Archive created successfully: files/c_files.zip
Archive contents:
 0  2025-09-23 06:37  c_files/
94  2025-09-23 06:37  c_files/hello.c
81  2025-09-23 06:37  c_files/utils.c

Creating archive: h_files.zip
Archive created successfully: files/h_files.zip
Archive contents:
 0  2025-09-23 06:37  h_files/
69  2025-09-23 06:37  h_files/utils.h
61  2025-09-23 06:37  h_files/main.h
```

Current directory structure:

```
files
├── c_files
│   ├── hello.c
│   └── utils.c
├── c_files.zip
├── data.json
├── h_files
│   ├── main.h
│   └── utils.h
├── h_files.zip
├── pdf_files
│   ├── document1.pdf
│   ├── manual.pdf
│   └── report.pdf
├── pdf_files.zip
└── readme.txt
```

4 directories, 12 files

Final contents of files:

```
total 32
drwxrwxrwx+ 5 codespace codespace 4096 Sep 23 06:37 .
drwxrwxrwx+ 5 codespace root      4096 Sep 23 06:37 ..
drwxrwxrwx+ 2 codespace codespace 4096 Sep 23 06:37 c_files
-rw-rw-rw-  1 codespace codespace  625 Sep 23 06:37 c_files.zip
-rw-rw-rw-  1 codespace codespace    0 Sep 23 06:37 data.json
drwxrwxrwx+ 2 codespace codespace 4096 Sep 23 06:37 h_files
-rw-rw-rw-  1 codespace codespace  579 Sep 23 06:37 h_files.zip
drwxrwxrwx+ 2 codespace codespace 4096 Sep 23 06:37 pdf_files
-rw-rw-rw-  1 codespace codespace  680 Sep 23 06:37 pdf_files.zip
-rw-rw-rw-  1 codespace codespace    0 Sep 23 06:37 readme.txt
```

File segregation and archiving completed successfully!

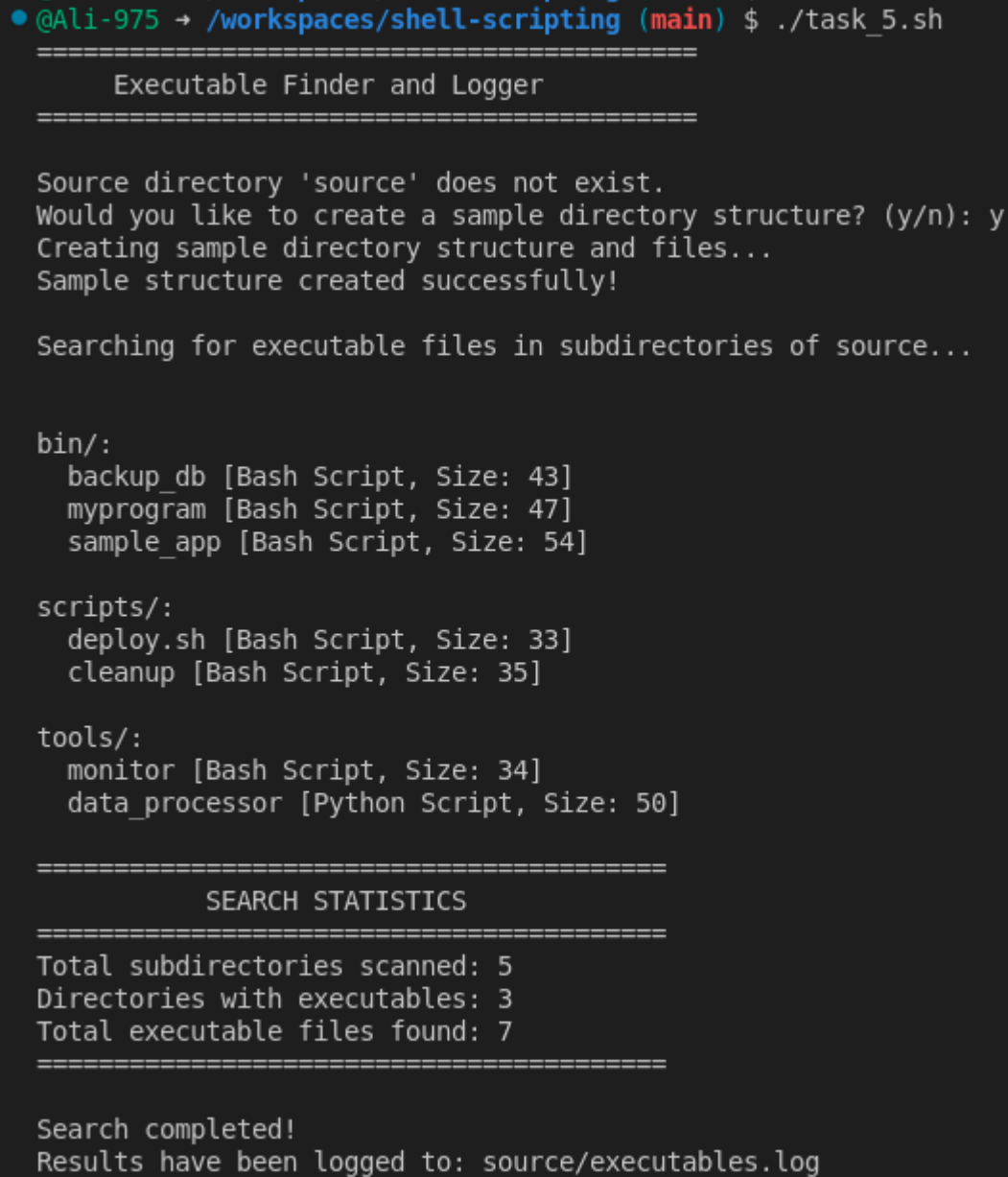
Summary:

- Files have been segregated by extension (.pdf, .c, .h)
- Separate subdirectories created for each file type
- Zip archives created for non-empty subdirectories
- All archives are stored in the files directory

```
▼ files
  > c_files
  > h_files
  > pdf_files
  ▣ c_files.zip
  {} data.json
  ▣ h_files.zip
  ▣ pdf_files.zip
  ⓘ readme.txt
```

v. Task 5:

The script recursively searches the `source/` directory for executable files. For every subdirectory, it lists the executables it contains and logs the same output to `source/executables.log`. This provides both on-screen results and a persistent record.



```
• @Ali-975 → /workspaces/shell-scripting (main) $ ./task_5.sh
=====
Executable Finder and Logger
=====

Source directory 'source' does not exist.
Would you like to create a sample directory structure? (y/n): y
Creating sample directory structure and files...
Sample structure created successfully!

Searching for executable files in subdirectories of source...

bin/:
  backup_db [Bash Script, Size: 43]
  myprogram [Bash Script, Size: 47]
  sample_app [Bash Script, Size: 54]

scripts/:
  deploy.sh [Bash Script, Size: 33]
  cleanup [Bash Script, Size: 35]

tools/:
  monitor [Bash Script, Size: 34]
  data_processor [Python Script, Size: 50]

=====
SEARCH STATISTICS
=====
Total subdirectories scanned: 5
Directories with executables: 3
Total executable files found: 7
=====

Search completed!
Results have been logged to: source/executables.log
```



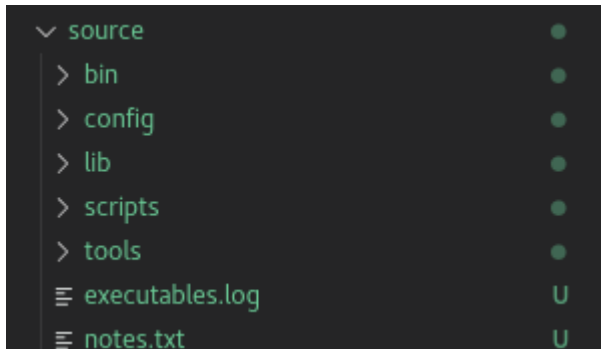
```
Log file preview:
=====
Executable Files Search Report
Generated on: Tue Sep 23 07:16:38 UTC 2025
Search Directory: /workspaces/shell-scripting/source
=====

bin/:
  backup_db [Bash Script, Size: 43]
  myprogram [Bash Script, Size: 47]
  sample_app [Bash Script, Size: 54]

scripts/:
  deploy.sh [Bash Script, Size: 33]
  cleanup [Bash Script, Size: 35]

tools/:
  monitor [Bash Script, Size: 34]
  data_processor [Python Script, Size: 50]

=====
... (truncated, see full log at source/executables.log)
@Ali-975 → /workspaces/shell-scripting (main) $
```



2. Critical Analysis:

In Lab 27, I implemented and tested five practical shell scripts. These tasks covered core Linux concepts such as sorting algorithms, random data generation, file and directory manipulation, backup and change detection, and recursive searching with logging. Together, they strengthened my scripting skills, improved my understanding of Git-like workflows, and demonstrated how to automate everyday system administration tasks efficiently.